

BEYOND THE HORIZON

SKYGAZE INDIA

Also Inside: Astronomical Events | Astrophotographs
Lunar Calendar | Galaxies | Dark Matter | Multiverse Theory

OCTOBER 2025 | VOL. 1, ISSUE 4

INTRODUCTION

Beyond the Horizon, Skygaze India's monthly publication, is meticulously curated to stimulate intellectual curiosity and foster deeper public engagement with astronomy and space science.

Each issue presents an amalgamation of celestial event highlights, exemplary astrophotography features, perspicacious articles, and exclusive content contributed by our community of astronomy enthusiasts. Our objective is to make space science both accessible and profoundly inspiring.

OUR TEAM

Founder & Editor in Chief

Harsh

Contributors

Mina Gerges, UAE

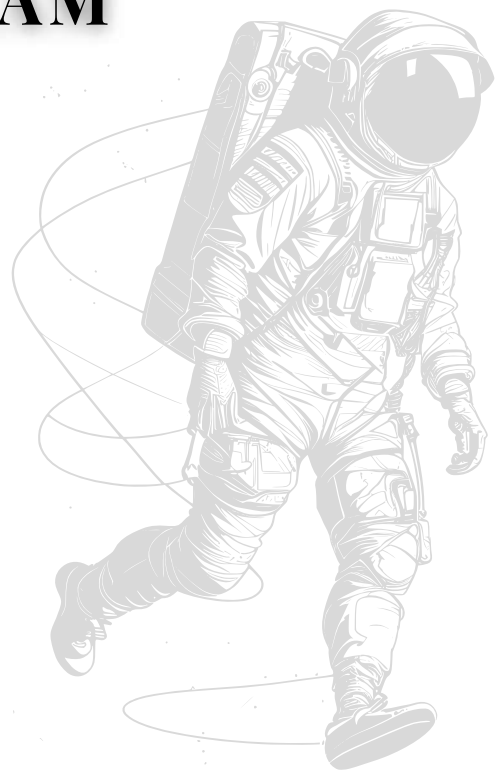
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"At Skygaze India, our mission is to make space science accessible—to bring the cosmos closer by turning complex knowledge into everyday curiosity for all."

Harsh
Founder & CEO

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ASTRONOMICAL EVENTS



OCTOBER 7: HUNTER'S FULL MOON

The Hunter's Moon, following the Harvest Moon, will rise in all its glory. Traditionally named as it provided extra moonlight for hunters preparing for winter, this full moon will appear especially bright and large in the night sky. A perfect opportunity for moon-gazers and astrophotographers alike.

OCTOBER 13: CONJUNCTION OF JUPITER AND THE MOON

A breathtaking sight awaits as Jupiter, the largest planet in our solar system, aligns closely with the Moon. Visible just after sunset, this conjunction offers a great chance to spot Jupiter's bright glow near the lunar disk with the naked eye—and even better through binoculars or a telescope.



OCTOBER 21: NEW MOON

The Moon will be in its new phase, making the night sky darker and ideal for stargazing. With minimal natural light, deep-sky objects like star clusters, nebulae, and even the Milky Way will be more visible from dark-sky locations. This is one of the best nights of the month for astronomers to enjoy uninterrupted views of the cosmos.

OCTOBER 21–22: ORIONIDS METEOR SHOWER

One of the reliable meteor showers, the Orionids, will peak on these nights. With the New Moon providing a dark backdrop, conditions will be excellent for observing meteors streaking across the sky at up to 20 per hour. The Orionids originate from Halley's Comet and appear to radiate from Orion.



LUNAR CALENDAR

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
			01  67.43% First Quarter 9.06 day	02  76.76% Waxing Gibbous 10.04 day	03  85.21% Waxing Gibbous 11.06 day	04  92.24% Waxing Gibbous 12.11 day
05  97.27% Full Moon 13.21 day	06  99.79% Full Moon 14.33 day	07  99.43% Full Moon 15.48 day	08  96.12% Waning Gibbous 16.63 day	09  90.08% Waning Gibbous 17.78 day	10  81.8% Waning Gibbous 18.91 day	11  71.92% Waning Gibbous 20.01 day
12  61.13% Last Quarter 21.09 day	13  50.08% Last Quarter 22.14 day	14  39.3% Last Quarter 23.16 day	15  29.27% Waning Crescent 24.16 day	16  20.36% Waning Crescent 25.13 day	17  12.84% Waning Crescent 26.09 day	18  6.94% Waning Crescent 27.02 day
19  2.8% New Moon 27.95 day	20  0.5% New Moon 28.86 day	21  0.06% New Moon 0.24 day	22  1.45% New Moon 1.13 day	23  4.57% Waxing Crescent 2.03 day	24  9.32% Waxing Crescent 2.92 day	25  15.52% Waxing Crescent 3.81 day
26  23.01% Waxing Crescent 4.7 day	27  31.57% First Quarter 5.61 day	28  40.98% First Quarter 6.53 day	29  50.97% First Quarter 7.47 day	30  61.22% First Quarter 8.45 day	31  71.32% Waxing Gibbous 9.45 day	



MIND-BLOWING FACT OF THE MONTH

Our Milky Way galaxy is on a collision course with the Andromeda Galaxy. The two are expected to merge in about 4.5 billion years, forming a giant elliptical galaxy astronomers call "*Milkomeda*". When they collide, stars themselves are unlikely to crash due to the vast distances between them.

However, their gravitational interaction will reshape both galaxies, sparking new waves of star formation and creating a spectacular cosmic transformation.

SEPTEMBER RECAP

STARGAZING AT RICHMOND SOCIETY

On 28th September 2025, Skygaze India organized an enchanting stargazing session at Richmond Society, Gurugram. Residents of all ages turned the community space into an open-air astronomy classroom.

Using our 10-inch Dobsonian telescope, participants explored the Moon, Saturn, and bright stars across the night sky — many seeing Saturn's majestic rings for the first time.

The excitement and curiosity made the evening truly memorable, with residents eagerly looking forward to future Skygaze sessions.

“We saw Saturn’s rings, and it was so exciting! I was really surprised by how beautiful the rings looked and how colorful they were. The Moon’s craters were amazing too! It was so cool to see one side bright from the Sun and the other side dark. Both looked really beautiful — and Saturn’s rings were the most beautiful of all!”

Children from Richmond Society



COLLABORATION WITH PACS E-LAB

Skygaze India has officially signed a Memorandum of Understanding (MoU) with the Pan-African Citizen Science e-Laboratory (PACS e-Lab), a leading nonprofit dedicated to advancing STEM and astronomy education across Africa. The agreement, signed on September 24th, 2025, marks the beginning of a meaningful partnership aimed at strengthening international cooperation in astronomy outreach, education, and citizen science.

Through this collaboration, both organizations reaffirm their shared commitment to making astronomy more inclusive, accessible, and globally connected. Together, Skygaze India and PACS e-Lab aspire to inspire curiosity about the cosmos, nurture young scientific minds, and promote cross-cultural learning between India and Africa.

This partnership stands as a testament to the unifying power of science — reminding us that while borders may divide us, the sky belongs to everyone.



FREE TOOLS BY SKYGAZE INDIA

Want to take your curiosity beyond the pages?

Explore the night sky with **free tools, guides, and resources from Skygaze India**. Discover telescopic tools, sky maps, and observation aids — all in one place:

- Planet Rise & Set Tracker
- ISS Live Tracker
- Astronomy Quiz
- Near-Earth Object Tracker
- Telescope Magnification Calculator



Discover Now

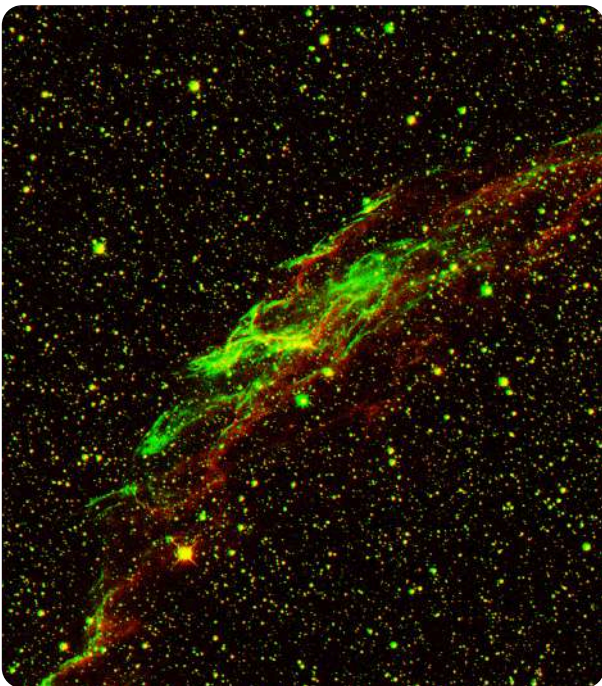
ASTROPHOTOGRAPHS



The Red Moon captured by Mr. Harsh, Founder at Skygaze India, Delhi, India



The Red Moon captured by Mr. Vishal, Amateur Astronomer, Delhi, India



Veil Nebula captured by Mr. Gukuu Justpa, Student at University of Energy and Natural Resources, Sunyani, Ghana



The Harvest Full Moon captured by Mr. Harsh, Founder at Skygaze India, Delhi, India

THE INVISIBLE GLUE OF THE UNIVERSE: DARK MATTER AND GALAXIES

If you look up at the night sky and see thousands of stars twinkling, it's easy to imagine that's what makes up most of the universe. Yet, astonishingly, everything we can see — stars, planets, dust, and gas — is just **a tiny fraction** of what's really out there. The rest of the universe is dominated by something invisible, mysterious, and utterly essential — **dark matter**.

Galaxies like our Milky Way are immense systems containing billions of stars rotating around a common center. But when astronomers began studying how these galaxies spin, they noticed something strange. The stars at the outer edges were moving **just as fast** as those near the center — a behavior that shouldn't be possible if only visible matter existed.

According to the laws of gravity, stars farther away should move slower, or else they'd fly off into space. Yet they didn't. This led scientists to realize there must be **something else** — a vast amount of unseen mass providing the extra gravity needed to hold galaxies together.

That mysterious mass was named **dark matter** — the invisible glue of the universe.

Discovering the Invisible

The first hints of dark matter go back to the 1930s, when astronomer **Fritz Zwicky** studied galaxy clusters and found they had far more mass than what could be seen. Later, in the 1970s, Vera Rubin confirmed this through her precise observations of galaxy rotation curves.

She discovered that galaxies behave as though surrounded by an unseen halo of matter extending far beyond their visible edges.

This invisible material doesn't emit, absorb, or reflect light — it can't be seen with telescopes of any kind. But its **gravitational effects** are undeniable.

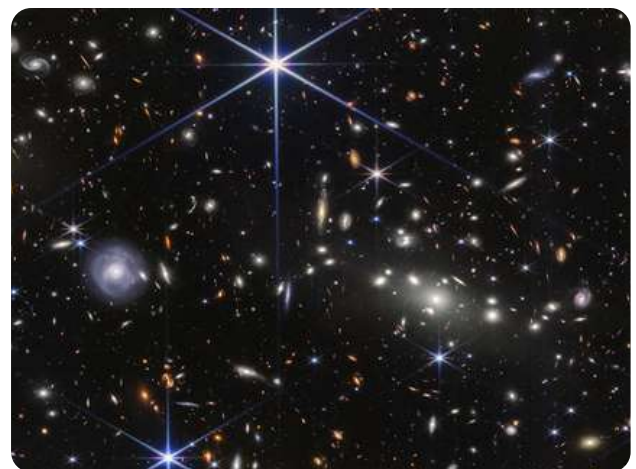
What Is Dark Matter Made Of?

That's the million-dollar question — and one of modern science's biggest mysteries.

Dark matter isn't made of ordinary atoms like us. It's not dust, planets, or black holes. Instead, scientists believe it could be made of exotic particles — perhaps WIMPs (Weakly Interacting Massive Particles) or axions — that rarely interact with normal matter.

Despite decades of searching, no one has detected these particles directly.

Yet, through their gravitational influence, dark matter reveals its presence everywhere — from the formation of galaxies to the large-scale structure of the entire universe.



The Bullet Cluster



A composite of Stephan's Quintet

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How Dark Matter Shapes Galaxies

Dark matter acts as a **cosmic framework** around which galaxies form and evolve.

Imagine dark matter as an enormous, invisible web stretching across the universe. In the densest regions of this web, gravity pulls in gas and dust, giving birth to galaxies and clusters. Each galaxy, including our own Milky Way, sits inside a vast **dark matter halo** — an unseen sphere that keeps it stable and prevents it from breaking apart as it spins.

Without dark matter, galaxies would never have formed the way they did — or might not have formed at all. It is truly the **architect of the cosmos**.

Searching for the Unseen

Around the world, scientists are racing to unlock the nature of dark matter.

- Deep underground detectors like **LUX-ZEPLIN** and **XENONnT** are built to catch rare collisions between dark matter and normal matter.
- The **Large Hadron Collider (CERN)** smashes particles together, hoping to create dark matter in controlled experiments.
- Telescopes and space missions, including **the James Webb Space Telescope** and **ESA's Euclid**, help trace dark matter's influence on galaxies and cosmic structure.

-Harsh

Founder, Skygaze India

ANOTHER VERSION OF YOU: A SCIENTIFIC VISION OF THE MULTIVERSE THEORY

The universe has always invited one profound question — are we truly alone?

This article explores one of modern science's most fascinating frontiers: the **multiverse theory**, the idea that our universe might be just one among countless others.

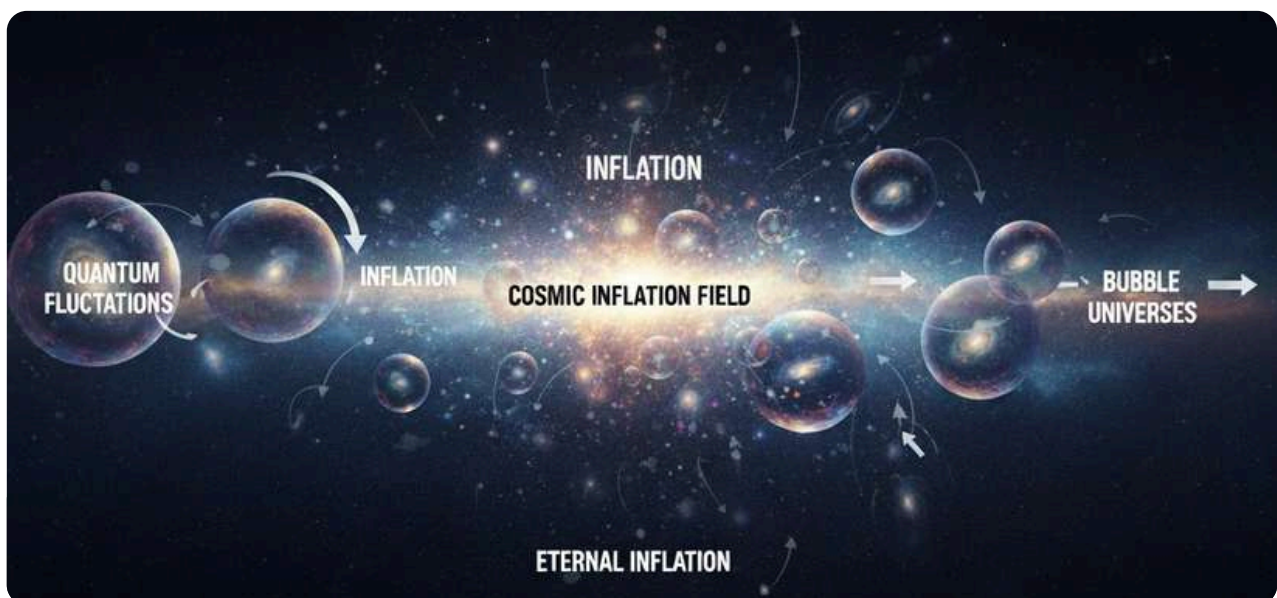
It begins by revisiting the scientific revolution of the twentieth century, when **quantum mechanics** replaced the predictable world of **Newton and Einstein** with one ruled by probability. Through phenomena such as **quantum entanglement** — what Einstein once called “spooky action at a distance” — we learn that the very fabric of reality can defy our common sense of space, time, and causality.

Building upon this quantum foundation, the discussion expands to a **cosmic scale**, connecting ideas such as **cosmic inflation** (the rapid expansion of the early universe) and **eternal inflation** (the possibility that this process never truly ended). These theories suggest that our universe may be just one bubble in an immense cosmic sea, each bubble potentially shaped by its own physical laws and constants.

Yet, while most scientific discussions of the multiverse focus on **physics**, this article turns to a more rarely explored dimension — **biology**.

Here lies the true originality of this work. Rather than treating the multiverse as an abstract physical concept, it asks a deeper question:

“Could life — and especially human life — ever truly be replicated across universes?”



Drawing on insights from **molecular biology, environmental science,** and **astrobiology**, the article argues that the emergence of life depends on a precise harmony of factors — chemical compositions, planetary environments, and evolutionary chance — that makes exact repetition nearly impossible.

Even if atoms were arranged identically elsewhere, the birth and growth of life rely on far more than molecular similarity.

This forms the **intellectual core** of the piece: a critique of the “other versions of us” idea.

Thinkers like **Brian Greene and Nick Bostrom** have proposed that in an infinite universe, identical copies of you and me could exist somewhere else. Yet this article exposes a biological flaw in that argument.

The complexity of DNA, its dependence on planetary conditions, and the unpredictable nature of evolution make such duplication extraordinarily unlikely.

Even tiny differences — in temperature, atmosphere, or chemical balance — could completely alter the course of evolution. The multiverse may indeed be infinite, but exact human replicas are probably not among its possibilities.

Recent research, such as *“Multiverse Predictions for Habitability: The Habitability of Exotic Environments”* (2025), supports this reasoning by suggesting that habitable conditions could exist in strange and unexpected forms across the cosmos. The discussion also aligns with **NASA’s Astrobiology Primer 3.0**, which urges scientists to look beyond Earth-like life and embrace a broader definition of what it means to be “alive.”

Thus, the most plausible vision of the multiverse is not one filled with duplicates of ourselves, but one teeming with **diverse and adaptive life forms**, each following its own unique rules of complexity and evolution.

In its closing reflection, the article bridges **science and philosophy**. It argues that truly understanding the multiverse requires more than equations — it demands **synthesis, imagination, and humility**. The author concludes with a timeless thought:

“Before being a physicist, be a philosopher. And before being a philosopher, be a thinker who masters the logic of synthesis, rationality, and precision.”

Ultimately, this piece does not just describe the multiverse — it reimagines it. It turns a speculative scientific idea into a **mirror of human curiosity**, reminding us that while the cosmos may be infinite, the search for meaning begins within — in our ability to question, reason, and wonder.

-Assil Bedra

Student, Khadraoui Brothers High School

GALAXIES: CLASSIFICATION AND PROPERTIES

Introduction

A galaxy is defined as a group of stars, clouds of dust and gas that are held together by gravitational force. To infer how do galaxies form, evolve and interact, astronomers study their different types (An introduction to Astrophysics, 2002). Therefore, the classification based on Morphological schemes remains fundamental. The first morphological scheme was introduced by Edwin Hubble (1920s-1930s), it's often called **"tuning-fork diagram"** or **"Hubble Sequence"**, they are classified into 3 main types: ellipticals, spirals, and irregulars. ("Galaxy - Spiral, Elliptical, Dwarf | Britannica," 2025)

Main Types and Their Properties

Here, we will introduce the main morphological types with their features in a comparative method in table 1.(Morison, 2009)

Galaxy Type	Morphological Range / Subclass	Structure & Shape	Star Population	Gas & Dust Content	Examples
Elliptical (E0–E7)	E0 = round → E7 = elongated	Smooth, oval or spherical; no spiral arms	Old, red stars	Very little gas or dust	M87 (Virgo-Cluster)
Spiral (Sa–Sc)	Sa: tight arms, large bulge → Sc: loose arms, small bulge	Flat disk with spiral arms and central bulge	Mix of old (bulge) and young (arms) stars	Rich in gas and dust	Milky Way, Andromeda (M31)
Barred Spiral (SBa–SBc)	SBa: tight arms, large bar → SBc: loose arms, small bar	Central bar of stars with spiral arms extending from ends	Old stars in bar; young stars in arms	Moderate to rich	Milky Way, NGC 1300
Lenticular (S0)	Transitional type between E and Sa	Disk and bulge; no spiral arms	Mainly old stars	Very little gas or dust	NGC 2787
Irregular (Irr I & II)	No defined Hubble subclass	No regular structure; chaotic	Mix but dominated by young stars	Gas-rich and dusty	Large & Small Magellanic Clouds

1. **Elliptical galaxies :**

Those galaxies are round or oval shaped with no visible structure like spiral arms. Those galaxies have existed in clusters (many galaxies together). (Kachelrieß, n.d.)

2. **Spiral Galaxies:**

According to Hubble's classification spiral galaxies are indicated by the following symbols as Sa, Sb, Sc based on how their arms are tightly wound and how large their bulge is.

3. **Barred spiral galaxies (SBa, SBb, SBc):**

Bar-shaped galaxy's structure, they form about 2/3 of spiral galaxies. The bar can channel gas leading to the formation of stars or feeding a central of a black hole (Buta et al., 2015).

4. **Lenticular Galaxies (S0):**

They are considered as intermediate between spiral and elliptical galaxies. They are also common in clusters. ("Galaxy - Spiral, Elliptical, Dwarf | Britannica," 2025)

5. Irregular Galaxies: They are characterized with no definite shape, rich in gas and forming star actively.

Galaxies and Evolution

Galaxies do not remain the same forever, their shape changes over time through different ways. When two galaxies collide together, this might lead to the formation of elliptical galaxies for spirals; this is called **Mergers**. Some other galaxies may lose their gas during a ram-pressure or supernova driven wind and transform to Lenticular (S0), this called **gas loss** (Masters, 2025). The gradual formation or dissolution of **Bars** due to the redistribution of material, which is called "**secular processes**". This confirms that galaxies are dynamic and continuously evolving.

Conclusion

The diversity of our universe structure is represented by the main four types: Elliptical, Spiral, Lenticular and Irregulars. Modern studies shows that the galaxy shape and structure depend on mass, environment and evolution. The understanding of how cosmos changes over time will be easier through studying galactic evolution.

-Mina Gerges

Department of Physics, College of Science, United Arab Emirates University

CONTRIBUTE TO BEYOND THE HORIZON!

Have an astronomy article, story, or thought to share?
We're inviting written submissions for our next publication!

Submit via email on support@skygazeindia.com



THIS MONTH IN INDIAN SPACE HISTORY

OCT, 2003

PSLV-C5 / RESOURCESAT-1 (IRS-P6) LAUNCH

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On 17 October 2003, ISRO's PSLV-C5 successfully launched the Earth observation satellite Resourcesat-1 (IRS-P6) from Sriharikota.

- Resourcesat-1 was the most advanced remote sensing satellite for India at that time, with a mass of around 1,360 kg.
- It was inserted into an 817 km polar Sun-synchronous orbit, continuing India's imaging and earth-observation capabilities.

OCT, 2008

LAUNCH OF CHANDRAYAAN-1

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On 22 October 2008, India launched its first lunar mission, Chandrayaan-1, from Satish Dhawan Space Centre, Sriharikota, using a PSLV-XL rocket.

- The mission carried 11 scientific instruments and aimed to map the Moon's surface and detect water molecules.
- Chandrayaan-1 deputed a Moon Impact Probe (MIP) in November 2008, which later confirmed the presence of water molecules near the lunar south pole.

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We're bringing the universe closer to you!

Join us at Select City Mall for an unforgettable skygazing experience — explore the sky through powerful telescopes, learn fascinating cosmic facts, and see the stars like never before.

Discover. Observe. Wonder.

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